

# Bastían Ibáñez Martínez

*Software Engineer · Reliable distributed systems & applied AI*

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## Profile

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Software engineer who designs AI agent platforms and the reliability machinery that keeps them running in production: exactly-once delivery, idempotency, concurrency control and traceability. Builds with explicit boundaries between domain, integration and infrastructure, and with recovery paths designed in from the start. Publishes libraries an entire organization depends on.

## Experience

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### Lead Software Engineer

*Aug. 2025 – Present*

*NotoriosTI – Santiago, Chile (Hybrid)*

- First engineering hire. Built the technical platform from scratch and was promoted to Lead as the team grew to three; owns architecture, GCP deployment strategy and technical direction.
- **Configuration and secrets library (Python + TypeScript, both published)** — Unifies `.env` and GCP Secret Manager under declarative YAML, with type coercion, multi-environment support, validation, secret masking and encrypted dotenv (ECIES/secp256k1, dotenvx-compatible) plus a CLI. Runtime dependency of 20+ repositories across the organization.
- **Multi-agent platform over ERP (LangGraph + FastAPI + Cloud Run)** — Designed the reliability machinery: transactional outbox with fencing tokens for exactly-once delivery, saga with fingerprint-based recovery over Odoo operations, two-layer conversation locks (Postgres advisory + asyncio) and backpressure coupled to model quota. ~1:1 test-to-code ratio, 40 ADRs, 15 idempotent migrations.
- **CI/CD and deployment security** — Workload Identity Federation with no long-lived keys, build-once with digest-pinned deploys and signed attestation, compare-and-swap traffic cutover on Cloud Run with automatic rollback, and a drift auditor for repository controls.
- **MercadoLibre–Odoo integration middleware** — FastAPI service that validates HMAC webhook signatures, maps products by SKU from PostgreSQL and creates sales orders in Odoo via RPC; OAuth token rotation with Secret Manager as the single source of truth.
- **Operational monitoring dashboard** — App tracking inventory, stockouts, sales and production, deployed on GCP (VM + Docker + systemd) with secrets from Secret Manager.
- *Stack: Python, TypeScript, LangGraph, FastAPI, PostgreSQL, BigQuery, Odoo, GCP (Cloud Run, Cloud SQL, Secret Manager, Cloud Tasks, WIF), Docker.*

### Junior Software Engineer

*May 2025 – Aug. 2025*

*NotoriosTI – Santiago, Chile (Hybrid)*

- Stood up the company's technical infrastructure from zero: full GCP setup (Cloud Run, Cloud SQL, Secret Manager, BigQuery), Docker pipelines and deployment strategy.
- Built the foundations of the LangGraph multi-agent assistant —agent architecture, Slack integration and Odoo connectivity— and **odoo-engine**, a published Python library encapsulating Odoo integration (JSON-RPC, models, CRUD).

Published Packages & Open Source

**env-manager**  
Configuration loader with GCP Secret Manager support, type coercion and aggregated error validation.  
github.com/NotoriosTI/env-manager

*PyPI*

**@notoriosTI/env-manager**  
TypeScript port with ECIES/secp256k1 encrypted dotenv and CLI; dual ESM+CJS build, strict typing.  
github.com/NotoriosTI/env-manager-js

*npm*

Personal Projects

**Manufacturing operations platform**  
Full-stack product: traceable inventory, transactional manufacturing, purchasing, quality, scrap and reporting. Next.js 16 + React 19 + Convex (serializable ACID mutations) + Clerk, on Cloudflare Workers. 120 test files, 1,749 assertions (Vitest + Playwright).

**go-agent**  
Terminal coding agent in Go, hexagonal architecture: ReAct core isolated from TUI, model providers, tools and persistence. Four model adapters, streaming, cancellation and confirmation for sensitive actions.

**GroupFit**  
Planner that assigns groups to hotels and dates under capacity, availability and transport constraints, with an explainable partial result. Heuristic with repair + optional ILP refinement (PuLP/HiGHS); FastAPI + Next.js.

**issue-agent**  
LangGraph triage agent: bug reports in Slack → routed GitHub issues with a suggested fix. Cloud Tasks as a database-free debounce buffer, on scale-to-zero Cloud Run with WIF.

Technical Skills

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|---------------|--------------------------------------------------------------------------------------------------------------------|
| Languages     | Python, TypeScript, SQL / PL-pgSQL                                                                                 |
| Backend       | FastAPI, LangGraph, LangChain, Fastify, LLM agents, RPC                                                            |
| Frontend      | React, Next.js (App Router), Convex, Tailwind                                                                      |
| Data          | PostgreSQL, BigQuery, pgvector, embeddings, forecasting (SARIMA)                                                   |
| Cloud / infra | GCP (Cloud Run, Cloud SQL, Secret Manager, Cloud Tasks, WIF, Artifact Registry), Docker, Cloudflare Workers, CI/CD |
| Testing       | pytest, Vitest, Playwright; integration tests against real Postgres                                                |
| Integrations  | Odoo (JSON-RPC / XML-RPC), Shopify, MercadoLibre, Slack, Chatwoot, WhatsApp Cloud API                              |

| Education                                                                                                               | Languages                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>BSc/MEng in Computer Engineering</b> 2022 – 2028 ( <i>exp.</i> )<br><i>Universidad Andrés Bello, Santiago, Chile</i> | <b>Spanish</b> Native<br><b>English</b> C1 – Cambridge FCE (2021)<br><b>German</b> C1 – DSD II, KMK (2021) |